A BESSEL INEQUALITY FOR POINT EVALUATIONS IN HARDY AND BERGMAN SPACES OF DIRICHLET SERIES

ØISTEIN SØVIK

ABSTRACT. We bound $\int |F|^p \, \mathrm{d}\mu$ on the Hardy space of the infinite polytorus, and on product weighted Bergman spaces of the infinite polydisc, by the norm of one positive integral operator on $L^2(\mu)$, for every $0 < p < \infty$ at once. Its kernel is the inner product of the moduli of two reproducing kernels, and the proof replaces the inner-outer factorization, which fails in infinitely many variables, by a sub-mean-value inequality for $|F|^{p/2}$ and Bessel's inequality. Consequences include Perfekt's Fejér–Riesz inequality for Dirichlet series for every p with the constant $5 + 2\sqrt{2}$, its Bergman counterpart, a bounded symbol for the Bergman multiplicative Hilbert matrix, and a characterization of Carleson measures on the curves $\sigma \mapsto (p^{-1/2-\sigma})_{p \in R}$ for sets of primes R with a density.

1. INTRODUCTION

Let $\mathscr{H}^p$ denote the Hardy space of Dirichlet series $f(s) = \sum_{n \geq 1} a_n n^{-s}$, which for $0 < p < \infty$ is the closure of the Dirichlet polynomials in the norm (quasi-norm if $p < 1$)

$$\|f\|_{\mathscr{H}^p} = \Big(\int_{\mathbb{T}^\infty} |F(z)|^p \, \mathrm{d}m_1^\infty(z) \Big)^{1/p},$$

where $F(z) = \sum_n a_n z^{\kappa(n)}$ is the Bohr lift of f, with $n = \prod_j p_j^{\kappa_j(n)}$, and m_1^∞ is the Haar measure of the infinite polytorus [2, 8]. For $\alpha > 1$ let m_α be the probability measure $(\alpha - 1)(1 - |z|^2)^{\alpha - 2} \, \mathrm{d}A(z)$ on $\mathbb{D}$, where dA is normalized area measure, and let $\mathscr{A}^p_\alpha$ be the corresponding space with $\mathbb{D}^\infty$ and m_α^∞ in place of $\mathbb{T}^\infty$ and m_1^∞. The case $\alpha = 2$ is product area measure, studied by Bailleul and Lefèvre [1]; we write $\mathscr{A}^p = \mathscr{A}^p_2$. We also write $\mathscr{A}^p_1 = \mathscr{H}^p$.

A point evaluation $f \mapsto f(s)$ is bounded on $\mathscr{H}^p$ exactly when $\operatorname{Re} s > 1/2$, by a theorem of Cole and Gamelin [5]. This paper is about how many point evaluations can be large at once, that is, about inequalities of the form

$$\int |f(1/2 + \sigma)|^p \, \mathrm{d}\nu(\sigma) \leq C \|f\|^p_{\mathscr{A}^p_\alpha}. \tag{1}$$

Date: October 12, 2026.

2020 *Mathematics Subject Classification.* Primary 30B50; Secondary 30H10, 30H20, 47B35.

Key words and phrases. Hardy spaces of Dirichlet series, Bergman spaces, Carleson measures, Fejér–Riesz inequality, multiplicative Hilbert matrix.

In one variable such inequalities are reduced to $p = 2$ by the factorization $F = IG^{2/p}$ with I inner and $G \in H^2$. In infinitely many variables this factorization fails, and there is no direct reduction to the case $p = 2$ unless $p \in 2\mathbb{N}$.

For $p = 2$ and Lebesgue measure on $(0, 1)$, (1) is the boundedness of the multiplicative Hilbert matrix of Brevig, Perfekt, Seip, Siskakis and Vukotić [4]. The case $p = 1$ was raised there, and restated by Saksman and Seip [10, Problem 3.2]: it is equivalent to the existence of a bounded symbol for the multiplicative Hilbert matrix. Perfekt [9] recently solved it. He proved that

$$\int_0^1 |f(1/2 + \sigma)| \, \mathrm{d}\sigma \le C\|f\|_{\mathscr{H}^1} \tag{2}$$

by localizing representatives of point evaluations and proving, through separation and concentration estimates, that the localized representatives have bounded overlap. Applying (2) to f^m gives the same inequality in $\mathscr{H}^m$ for every positive integer m. In the Bergman setting, Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [3] introduced the analogue of the multiplicative Hilbert matrix, the form on $\mathscr{A}^2 \times \mathscr{A}^2$ with analytic symbol

$$\varphi(s) = \sum_{n \ge 2} \frac{d(n)}{\sqrt{n} \log^2 n} \, n^{-s},$$

proved that it is bounded and not compact, and that φ defines a bounded functional on $\mathscr{A}^p$ for $1 < p < \infty$ [3, Theorems 13 and 16].

The Hellinger–Bessel inequality. The main purpose of the present paper is to prove a single inequality from which these estimates follow for every $0 < p < \infty$, with explicit constants. Fix $\alpha \ge 1$ and work in finitely many variables $z = (z_1, \dots, z_d)$, on $\Omega = \mathbb{T}^d$ if $\alpha = 1$ and on $\Omega = \mathbb{D}^d$ if $\alpha > 1$, with the product measure $m = m_\alpha^d$. The reproducing kernel of the analytic polynomials in $L^2(m)$ is

$$K_r(z) = \prod_{\ell=1}^{d} (1 - \overline{r_\ell} z_\ell)^{-\alpha}, \qquad \|K_r\|_2^2 = K_r(r) = \prod_{\ell=1}^{d} (1 - |r_\ell|^2)^{-\alpha},$$

for $r \in \mathbb{D}^d$. We write $h_r = |K_r|/\|K_r\|_2$, a positive unit vector in $L^2(m)$, and $\omega_r = h_r^2 \, m$ for the corresponding probability measure. For $\alpha = 1$ it is the harmonic measure of r; in general it is the image of m under a Möbius map (Lemma 1). The inner product $\langle h_r, h_s \rangle$ is the Hellinger affinity of ω_r and ω_s. For a positive Borel measure μ on $\mathbb{D}^d$, let $\mathcal{G}_\mu$ be the integral operator on $L^2(\mu)$ with the positive kernel

$$\mathcal{G}(r, s) = \int |K_r||K_s| \, \mathrm{d}m = \|K_r\|_2 \|K_s\|_2 \langle h_r, h_s \rangle.$$

The two factors separate the size of a point evaluation, $\|K_r\|_2^2$, from the overlap of two normalized evaluation profiles, $\langle h_r, h_s \rangle$.

Theorem 1. *Suppose that $\alpha \geq 1$, that μ is a positive Borel measure on $\mathbb{D}^d$, and that $\mathcal{G}_\mu$ is bounded on $L^2(\mu)$. Then for every $0 < p < \infty$ and every analytic polynomial F,*

$$\int |F|^p \, \mathrm{d}\mu \leq \|\mathcal{G}_\mu\| \int |F|^p \, \mathrm{d}m.$$

The constant depends neither on p nor on d. For $\mu = \delta_r$ the theorem is the estimate $|F(r)|^p \leq \|K_r\|_2^2 \|F\|_p^p$ of Cole and Gamelin, with equality for $F = K_r^{2/p}$. Since $\mathcal{G}(r,s) \geq |\langle K_r, K_s \rangle|$, the condition of Theorem 1 is stronger than boundedness of the operator with kernel $\langle K_r, K_s \rangle$, which characterizes (1) for $p = 2$; this is the price of passing from K_r to $|K_r|$.

Consequences. Along the curve $\sigma \mapsto 1/2 + \sigma$ the natural measure for $\mathscr{A}^p_\alpha$ is $\sigma^{\alpha-1} \, \mathrm{d}\sigma$.

Theorem 2. *Suppose that $\alpha \geq 1$ and $0 < p < \infty$. Then every $f \in \mathscr{A}^p_\alpha$ satisfies*

$$\int_0^1 |f(1/2+\sigma)|^p \, \sigma^{\alpha-1} \, \mathrm{d}\sigma \leq C_\alpha \|f\|^p_{\mathscr{A}^p_\alpha}, \qquad C_\alpha = \Big(\frac{3}{\sqrt{2}}\Big)^\alpha B\Big(\frac{\alpha}{4}, \frac{\alpha}{4}\Big).$$

Moreover, C_1 can be replaced by $5+2\sqrt{2}$, and C_2 by $(\pi+5/2+1/\sqrt{3})/2 < 3.11$.

For $\alpha = 1$ and $p = 1$ this is Perfekt's inequality (2) with an explicit constant. Since $\sum_{n\geq 2} a_n/(\sqrt{n}\log n) = \int_0^\infty (f(1/2+\sigma) - a_1) \, \mathrm{d}\sigma$ for Dirichlet polynomials, we obtain the following.

Corollary 1. *For every $f \in \mathscr{H}^1$ the series $\sum_{n\geq 2} a_n/(\sqrt{n}\log n)$ converges, and*

$$\Big| a_1 + \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log n} \Big| \leq \Big(5 + 2\sqrt{2} + \sum_{n\geq 2} \frac{1}{n^{3/2}\log n}\Big) \|f\|_{\mathscr{H}^1} < 9\|f\|_{\mathscr{H}^1}.$$

Consequently the multiplicative Hilbert matrix has a bounded symbol B, in the sense of [9, Corollary 1.2]*, with $\|B\|_\infty < 9$.*

In the Bergman space $\mathscr{A}^1$ the same argument gives the endpoint of the result of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt.

Theorem 3. *For every $f \in \mathscr{A}^1$ the series $\sum_{n\geq 2} a_n/(\sqrt{n}\log^2 n)$ converges, and*

$$\Big| \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log^2 n} \Big| \leq 25 \|f\|_{\mathscr{A}^1}.$$

Consequently there is $\psi \in L^\infty(m_2^\infty)$ with $\|\psi\|_\infty \leq 25$ whose analytic projection is φ, that is, the Bergman multiplicative Hilbert form has a bounded symbol.

The constant 25 is not optimized; it is dominated by a crude bound for the part of the integral where $\sigma > 1$.

The last consequence shows that the condition of Theorem 1 loses nothing on the curves $\sigma \mapsto (p^{-1/2-\sigma})_{p\in R}$. A set R of primes has *density* $\theta \in (0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta \log\frac{1}{t}\Big| \le B \qquad (0 < t \le 2). \tag{3}$$

All primes have density 1, and by the prime number theorem in arithmetic progressions, a union of k reduced residue classes modulo q has density k/h, where h is the number of reduced residue classes modulo q. Let $r_R(\sigma) = (p^{-1/2-\sigma})_{p\in R}$, and for a positive measure ν on $(0,1]$ and $\kappa > 0$ put $[\nu]_\kappa = \sup_{0<t\le 1} t^{-\kappa}\nu((0,t])$.

Theorem 4. *Suppose that $\alpha \ge 1$, that R has density $\theta \in (0,1]$ with constant B, and that ν is a positive measure on $(0,1]$. Put $\kappa = \alpha\theta$. Then the following are equivalent:*

1. *there are $0 < p < \infty$ and $C < \infty$ such that $\int |F(r_R(\sigma))|^p \,\mathrm{d}\nu(\sigma) \le C \int |F|^p \,\mathrm{d}m_\alpha^\infty$ for every polynomial F in the variables indexed by R;*
2. *the same holds for every $0 < p < \infty$, with a constant independent of p;*
3. *$[\nu]_\kappa < \infty$;*
4. *for every finite $R' \subset R$, the operator $\mathcal{G}_{\nu,R'}$ on $L^2(\nu)$ with kernel $\mathcal{G}(r_{R'}(\sigma), r_{R'}(\tau))$, computed in the variables R', is bounded, with a bound independent of R'.*

Moreover, the best constant in (1) is comparable to $[\nu]_\kappa$, with constants depending only on α, θ and B.

To the best of our knowledge, Theorem 1, Theorem 2 for $\alpha > 1$, and Theorems 3 and 4 are new. For $\alpha = 1$, Theorem 2 is new when p is not an integer; for positive integers p the inequality with an unspecified constant follows from (2) as explained above, and for even p the case $p = 2$ gives a smaller constant. What is new for $\alpha = 1$ is therefore a single explicit constant for all p.

Our approach. Let us briefly describe our approach. The only place where analyticity enters is the sub-mean-value inequality

$$|F(r)|^q \le \int |F|^q |K_r| \,\mathrm{d}m \qquad (q > 0),$$

which follows from subharmonicity after the Möbius change of variables that Perfekt also uses. It is the mean-value argument behind the estimate of Cole and Gamelin [5], and since $|K_r|$ is a product it holds in any number of variables. With $q = p/2$ the function $u = |F|^{p/2}$ is a vector in $L^2(m)$ with $\|u\|_2^2 = \|F\|_p^p$, and the inequality reads $|F(r)|^p \le \|K_r\|_2^2 \langle u, h_r\rangle^2$. So the fraction of the Cole–Gamelin bound that F attains at r is at most the squared correlation of the unit vector $u/\|u\|_2$ with h_r. One unit vector cannot correlate strongly with many nearly orthogonal unit vectors, and Bessel's inequality makes this quantitative: the total correlation is bounded by the

Gram operator of the vectors $|K_r|$, which is $\mathcal{G}_\mu$. In contrast to Perfekt's argument, no localization is needed: boundedness of this Gram operator replaces bounded overlap of localized representatives, and the overlaps $\langle h_r, h_s\rangle$ can be computed exactly, one coordinate at a time.

Organization. Sections 2 and 3 contain the sub-mean-value inequality, the proof of Theorem 1, and the computation of the affinities. Theorem 2 is proved in Sections 4 and 5, Corollary 1 and Theorem 3 in Section 6, and Theorem 4 in Section 7. Section 8 contains concluding remarks.

2. Preliminaries

Throughout, $\alpha \geq 1$ and $d \geq 1$ are fixed, Ω and $m = m_\alpha^d$ are as above, and $\|F\|_p^p = \int |F|^p \,\mathrm{d}m$. For $r \in \mathbb{D}^d$ let T_r be the automorphism of $\mathbb{D}^d$ given by

$$T_r(w)_\ell = \frac{r_\ell + w_\ell}{1 + \overline{r_\ell} w_\ell},$$

which maps Ω onto itself and satisfies $T_r(0) = r$. Let $\rho(a,b) = |a-b|/|1-\overline{a}b|$ be the pseudo-hyperbolic distance on $\mathbb{D}$. Note that K_r is holomorphic and zero-free in a neighbourhood of the closed polydisc, since $|r_\ell| < 1$.

Lemma 1. *Suppose that $r \in \mathbb{D}^d$. Then ω_r is the image of m under T_r, that is,*

$$\int h \,\mathrm{d}\omega_r = \int h \circ T_r \,\mathrm{d}m$$

for every bounded Borel function h on Ω. Consequently $\int |H|^q \,\mathrm{d}\omega_r \geq |H(r)|^q$ for every $q > 0$ and every function H holomorphic in a neighbourhood of the closed polydisc.

Proof. Both measures are products, so we may assume that $d = 1$ and write $a = r$. For $\alpha = 1$ the Jacobian of T_a on $\mathbb{T}$ is $(1-|a|^2)/|1+\overline{a}w|^2$. For $\alpha > 1$, the identities $1 - |T_a w|^2 = (1-|a|^2)(1-|w|^2)|1+\overline{a}w|^{-2}$ and $|T_a'(w)|^2 = (1-|a|^2)^2|1+\overline{a}w|^{-4}$ show that the density of $m_\alpha \circ T_a$ with respect to m_α is $(1-|a|^2)^\alpha |1+\overline{a}w|^{-2\alpha}$. In both cases, since $K_a(T_a w) = ((1+\overline{a}w)/(1-|a|^2))^\alpha$, this density equals $\|K_a\|_2^2 |K_a(T_a w)|^{-2}$, and the substitution $z = T_a w$ gives

$$\int h \,\mathrm{d}\omega_a = \int h(z) \frac{|K_a(z)|^2}{\|K_a\|_2^2} \,\mathrm{d}m(z) = \int h(T_a w) \,\mathrm{d}m(w).$$

For the second claim, $\lambda \mapsto |H(T_r(\lambda w))|^q$ is subharmonic on a neighbourhood of the closed unit disc for every $w \in \Omega$, and m is invariant under the rotations $w \mapsto e^{i\theta} w$. Averaging over θ gives $\int |H \circ T_r|^q \,\mathrm{d}m \geq |H(T_r(0))|^q = |H(r)|^q$. $\square$

The next result is the sub-mean-value inequality described in the introduction. It is the only place where we use that F is analytic.

Lemma 2. *Suppose that $q > 0$, that F is an analytic polynomial, and that $r \in \mathbb{D}^d$. Then*

$$|F(r)|^q \leq \int |F|^q |K_r| \,\mathrm{d}m.$$

Proof. Let $K_r^{-1/q} = \prod_\ell (1-\overline{r_\ell} z_\ell)^{\alpha/q}$ with the principal branch, which is holomorphic near the closed polydisc since $\operatorname{Re}(1-\overline{r_\ell} z_\ell) > 0$ there. Then $H = FK_r^{-1/q}$ is holomorphic there. As $|F|^q |K_r| = \|K_r\|_2^2 |H|^q h_r^2$, Lemma 1 gives

$$\int |F|^q |K_r| \,\mathrm{d}m = \|K_r\|_2^2 \int |H|^q \,\mathrm{d}\omega_r \geq \|K_r\|_2^2 |H(r)|^q = |F(r)|^q,$$

where we used that $K_r(r) = \|K_r\|_2^2$. □

We will use Lemma 2 with $q = p/2$, which makes $|F|^q$ a vector in $L^2(m)$ whose squared norm is exactly $\|F\|_p^p$. There is equality for $F = K_r^{1/q}$, and the Cauchy–Schwarz inequality recovers the estimate of Cole and Gamelin.

3. The Hellinger–Bessel inequality

Let A be the analysis operator $(Au)(r) = \langle u, |K_r| \rangle$, from $L^2(m)$ to functions on $\mathbb{D}^d$. Its formal adjoint is $A^* c = \int c(s) |K_s| \,\mathrm{d}\mu(s)$, and $AA^* = \mathcal{G}_\mu$.

Proof of Theorem 1. For bounded c with compact support, Fubini's theorem gives $\|A^*c\|_{L^2(m)}^2 = \langle \mathcal{G}_\mu c, c \rangle_{L^2(\mu)} \leq \|\mathcal{G}_\mu\| \|c\|_{L^2(\mu)}^2$. Hence A^* extends to a bounded operator from $L^2(\mu)$ to $L^2(m)$ with $\|A^*\|^2 \leq \|\mathcal{G}_\mu\|$, and its adjoint is A, since $\langle u, A^*c \rangle = \int (Au) c \,\mathrm{d}\mu$ and Au is continuous. So

$$\int |Au|^2 \,\mathrm{d}\mu \leq \|\mathcal{G}_\mu\| \|u\|_2^2$$

for every $u \in L^2(m)$. Apply this to $u = |F|^{p/2}$. By Lemma 2 with $q = p/2$, $|F(r)|^p \leq (Au)(r)^2$, and $\|u\|_2^2 = \|F\|_p^p$. □

Remark 1. Two other routes seem natural, and neither works. The reproducing identity $F(r) = \langle F, K_r \rangle$ requires an analytic vector, and $F^{p/2}$ is in general not a holomorphic function unless $p \in 2\mathbb{N}$. The sub-mean-value inequality $|F(r)|^q \leq \int |F|^q \,\mathrm{d}\omega_r$ with respect to ω_r itself leads, with $q = p/2$, to the Gram kernel $\int (\mathrm{d}\omega_r/\,\mathrm{d}m)(\mathrm{d}\omega_s/\,\mathrm{d}m)\,\mathrm{d}m$ in place of $\mathcal{G}$; for $\alpha = 1$ along the curve $\sigma \mapsto (p^{-1/2-\sigma})_p$ this kernel equals $\zeta(1+\sigma+\tau)^2/\zeta(2+2\sigma+2\tau)$, which is of order $(\sigma+\tau)^{-2}$ near the origin and is unbounded on $L^2(0,1)$. With $q = p$ it gives $\int |F|^p \,\mathrm{d}\mu \leq \|\int (\mathrm{d}\omega_r/\,\mathrm{d}m)\,\mathrm{d}\mu(r)\|_\infty \|F\|_p^p$, and the Poisson kernels $\prod_p (1+p^{-1/2-\sigma})/(1-p^{-1/2-\sigma})$ at $z = 1$ are unbounded as the number of variables grows. The square root $|K_r| = \|K_r\|_2 (\mathrm{d}\omega_r/\,\mathrm{d}m)^{1/2}$ balances the two sides of the Cauchy–Schwarz inequality.

In applications, $\|\mathcal{G}_\mu\|$ is estimated by Schur's test: for every positive function w on the support of μ,

$$\|\mathcal{G}_\mu\| \leq \sup_r \frac{1}{w(r)} \int \mathcal{G}(r,s) w(s) \,\mathrm{d}\mu(s). \tag{4}$$

To use it we need the affinities $\langle h_r, h_s \rangle$. Both h_r and m are products, so $\langle h_r, h_s \rangle$ is the product of the one-variable affinities, and by Möbius invariance these depend only on the pseudo-hyperbolic distance.

Lemma 3. *Suppose that $\alpha \geq 1$ and $a, b \in \mathbb{D}$, and put $x = \rho(a,b)^2$. Then in one variable*

$$\langle h_a, h_b\rangle = (1-x)^{\alpha/2}\, {}_2F_1\Big(\frac{\alpha}{2}, \frac{\alpha}{2}; \alpha; x\Big) \leq (1-x)^{\alpha/4}.$$

Proof. Put $c = (b-a)/(1-\overline{a}b)$, so that $|c|^2 = x$. A direct computation gives

$$1 - \overline{b}\, T_a(w) = \frac{(1-a\overline{b})(1-\overline{c}w)}{1+\overline{a}w},$$

so $|K_b(T_a w)|/|K_a(T_a w)| = (1-|a|^2)^{\alpha}|1-\overline{a}b|^{-\alpha}|1-\overline{c}w|^{-\alpha}$. By Lemma 1,

$$\begin{aligned}\langle h_a, h_b\rangle &= \frac{\|K_a\|_2}{\|K_b\|_2}\int \frac{|K_b|}{|K_a|}\,\mathrm{d}\omega_a \\ &= \frac{(1-|a|^2)^{\alpha/2}(1-|b|^2)^{\alpha/2}}{|1-\overline{a}b|^{\alpha}}\int |1-\overline{c}w|^{-\alpha}\,\mathrm{d}m(w).\end{aligned}$$

The first factor is $(1-x)^{\alpha/2}$. Expanding $(1-\overline{c}w)^{-\alpha/2}$ in powers of w and using $\int |w|^{2k}\,\mathrm{d}m_\alpha = k!/(\alpha)_k$, we find that the integral is ${}_2F_1(\alpha/2, \alpha/2; \alpha; x)$. For the inequality, put $\beta = \alpha/2$. The power series of ${}_2F_1(\beta,\beta;2\beta;x)$ and of $(1-x)^{-\beta/2}$ both have constant term 1, and the ratios of their consecutive coefficients are $(\beta+k)^2/((2\beta+k)(k+1))$ and $(\beta/2+k)/(k+1)$. Since $(2\beta+k)(\beta/2+k) - (\beta+k)^2 = \beta k/2 \geq 0$, the first series is dominated coefficientwise by the second. □

The exponent $\alpha/4$ is the largest possible in a bound of the form $\langle h_a, h_b\rangle \leq (1-x)^{\gamma}$, since the identity in Lemma 3 gives $\langle h_a, h_b\rangle = 1 - \alpha x/4 + O(x^2)$ as $x \to 0$. On the other hand $|\langle K_a, K_b\rangle|/(\|K_a\|_2\|K_b\|_2) = (1-x)^{\alpha/2}$, so in one variable, and hence in d variables,

$$\frac{|\langle K_r, K_s\rangle|}{\|K_r\|_2\|K_s\|_2} \leq \langle h_r, h_s\rangle \leq \Big(\frac{|\langle K_r, K_s\rangle|}{\|K_r\|_2\|K_s\|_2}\Big)^{1/2}.$$

Taking products over the coordinates and using $1 - x \leq e^{-x}$, we obtain

$$\mathcal{G}(r,s) \leq \|K_r\|_2\|K_s\|_2 \exp\Big(-\frac{\alpha}{4}E(r,s)\Big), \qquad E(r,s) = \sum_{\ell=1}^{d}\rho(r_\ell, s_\ell)^2. \tag{5}$$

No single coordinate needs to separate r from s: the squared pseudo-hyperbolic distances add up, while the affinities multiply.

Along the prime curve, Lemma 3 takes a particularly simple form. Let $p_1 < p_2 < \dots$ be the primes, and for $\sigma > 0$ put $r(\sigma) = (p_\ell^{-1/2-\sigma})_{\ell \leq d}$, so that $F(r(\sigma)) = f(1/2+\sigma)$ for a Dirichlet polynomial f in the first d primes. With $a = p^{-1/2-\sigma}$ and $b = p^{-1/2-\tau}$ we have $1 - |a|^2 = 1 - p^{-1-2\sigma}$ and $1 - ab = 1 - p^{-1-\sigma-\tau}$, and the bound of Lemma 3 for one coordinate is $\|K_a\|_2\|K_b\|_2(1-x)^{\alpha/4} = ((1-a^2)(1-b^2))^{-\alpha/4}(1-ab)^{-\alpha/2}$. Each such factor is at least 1, so, uniformly in d,

$$\mathcal{G}\big(r(\sigma), r(\tau)\big) \leq \big(\zeta(1+2\sigma)\zeta(1+2\tau)\big)^{\alpha/4}\zeta(1+\sigma+\tau)^{\alpha/2}. \tag{6}$$

For $\alpha = 1$ the corresponding kernel for $p = 2$ is $|\langle K_{r(\sigma)}, K_{r(\tau)}\rangle| \le \zeta(1+\sigma+\tau)$, whose leading part $1/(\sigma+\tau)$ is Carleman's kernel, of norm π. The square root in the comparison above turns it into $2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$, of norm $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$; this is the loss introduced by the kernel comparison. We will use the elementary bound $\zeta(1+x) \le 1 + 1/x$ for $x > 0$.

4. The Fejér–Riesz inequality

Boundedness in Theorem 2 comes from homogeneity, which the following standard lemma records.

Lemma 4. *Suppose that k is a positive function on $(0,\infty)^2$ which is homogeneous of degree -1. Then the integral operator with kernel k on $L^2(0,\infty)$ has norm at most $\int_0^\infty k(1,x)x^{-1/2}\,\mathrm{d}x$.*

Proof. Schur's test with the weight $\tau^{-1/2}$ and the substitution $\tau = \sigma x$ give $\int_0^\infty k(\sigma,\tau)\tau^{-1/2}\,\mathrm{d}\tau = \sigma^{-1/2}\int_0^\infty k(1,x)x^{-1/2}\,\mathrm{d}x$. □

Proof of Theorem 2, with the constant C_α. Let f be a Dirichlet polynomial in the first d primes and let μ be the image of $\nu = \sigma^{\alpha-1}\,\mathrm{d}\sigma$ on $(0,1)$ under $\sigma \mapsto r(\sigma)$. By Theorem 1 it suffices to bound the norm of the kernel $\mathcal{G}(r(\sigma), r(\tau))$ on $L^2(\nu)$. The map $c \mapsto \sigma^{(\alpha-1)/2}c$ is unitary from $L^2(\nu)$ onto $L^2(0,1)$ and conjugates this operator to the one with kernel $(\sigma\tau)^{(\alpha-1)/2}\mathcal{G}(r(\sigma),r(\tau))$. By (6), $\zeta(1+x) \le (1+x)/x$, and $(1+2\sigma)^{1/4}(1+2\tau)^{1/4} \le (1+\sigma+\tau)^{1/2}$, this kernel is at most

$$M(\sigma,\tau) = 2^{-\alpha/2}(1+\sigma+\tau)^{\alpha}\,(\sigma\tau)^{\alpha/4-1/2}(\sigma+\tau)^{-\alpha/2}. \tag{7}$$

On $(0,1)^2$ we have $(1+\sigma+\tau)^\alpha \le 3^\alpha$, and the kernel $(\sigma\tau)^{\alpha/4-1/2}(\sigma+\tau)^{-\alpha/2}$ is homogeneous of degree -1. By Lemma 4 its norm is at most

$$\int_0^\infty x^{\alpha/4-1}(1+x)^{-\alpha/2}\,\mathrm{d}x = B\Big(\frac{\alpha}{4},\frac{\alpha}{4}\Big).$$

This proves the inequality with the constant C_α for Dirichlet polynomials. For general $f \in \mathscr{A}^p_\alpha$, the Cole–Gamelin estimate $|f(1/2+\sigma)|^p \le \zeta(1+2\sigma)^\alpha \|f\|^p_{\mathscr{A}^p_\alpha}$ shows that $f_k(1/2+\sigma) \to f(1/2+\sigma)$ for every $\sigma > 0$ when $f_k \to f$ in $\mathscr{A}^p_\alpha$, and Fatou's lemma gives the claim. □

5. The refined constants

In this section we prove the two refinements in Theorem 2. By the proof above, it suffices to bound the norm on $L^2(0,1)$ of a positive kernel which dominates $(\sigma\tau)^{(\alpha-1)/2}\mathcal{G}(r(\sigma),r(\tau))$.

The case $\alpha = 2$. Here (7) is

$$M(\sigma,\tau) = \frac{1}{2}\Big(\frac{1}{\sigma+\tau} + 2 + \sigma + \tau\Big).$$

On $L^2(0,1)$ the first kernel has norm at most π by Hilbert's inequality [7, Theorem 323], the constant kernel 2 has norm 2, and the kernel $\sigma+\tau$ has rank two and norm $1/2+1/\sqrt{3}$. This gives the constant $(\pi+5/2+1/\sqrt{3})/2$.

The case $\alpha=1$. By (6) and $\zeta(1+x)\le 1+1/x$, the kernel is dominated by

$$\overline{\mathcal{G}}(\sigma,\tau)=\Bigl(1+\frac{1}{2\sigma}\Bigr)^{1/4}\Bigl(1+\frac{1}{2\tau}\Bigr)^{1/4}\Bigl(1+\frac{1}{\sigma+\tau}\Bigr)^{1/2}.$$

Split $(0,1)=I_0\cup I_1$ with $I_0=(0,1/10)$ and $I_1=[1/10,1)$, and let $q(s)=1+(2s)^{-1/2}$. We bound the four blocks separately.

The block $I_0\times I_0$. Here $\overline{\mathcal{G}}(\sigma,\tau)\le (6/5)\,2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$. By Lemma 4 the norm of $2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$ is at most $2^{-1/2}B(1/4,1/4)=\Gamma(1/4)^2/\sqrt{2\pi}<5.25$, so this block has norm less than $(6/5)\cdot 5.25<7$.

The blocks $I_0\times I_1$ and $I_1\times I_0$. Since $(1+x)^{1/2}\le 1+x^{1/2}$, we have $\overline{\mathcal{G}}(s,t)^2\le q(s)q(t)(1+1/(s+t))$, and for $s\in I_0$, $t\in I_1$ we use $1/(s+t)\le 1/t$. Since $\int_{I_0} q\,\mathrm{d}s=1/10+1/\sqrt{5}$ and $\int_{I_1} q(t)(1+1/t)\,\mathrm{d}t=9/10+\log 10+\sqrt{20}-1/\sqrt{5}$, the square of the Hilbert–Schmidt norm is at most the product of these two numbers, which is less than 4, and the norm is less than 2.

The block $I_1\times I_1$. Here we use $1/(s+t)\le 1/(2\sqrt{st})$, which gives $\overline{\mathcal{G}}(s,t)^2\le q(s)q(t)+q(s)s^{-1/2}q(t)t^{-1/2}/2$. The square of the Hilbert–Schmidt norm is at most $A^2+C^2/2$, where $A=\int_{I_1} q=9/10+\sqrt{2}-1/\sqrt{5}$ and $C=\int_{I_1} q(s)s^{-1/2}\,\mathrm{d}s=2-2/\sqrt{10}+2^{-1/2}\log 10$. So it is less than 8, and the norm is less than 3.

Hence the norm of $\overline{\mathcal{G}}$ is at most the largest eigenvalue of the symmetric matrix with rows $(7,2)$ and $(2,3)$, which is $5+2\sqrt{2}$.

6. Summability and the multiplicative Hilbert forms

Theorem 2 gives bounded functionals on $\mathscr{H}^1$ and $\mathscr{A}^1$. That their values are given by the coefficient series in their natural order is a separate Tauberian matter, which we settle once for both spaces. We use two inequalities of Helson type. For $f\in\mathscr{H}^1$, Helson [6] proved that

$$\sum_{n\ge 1}\frac{|a_n|^2}{d(n)}\le\|f\|_{\mathscr{H}^1}^2, \tag{8}$$

and for $f\in\mathscr{A}^1$, Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [3, Theorem 10] proved that

$$\sum_{n\ge 1}\frac{|a_n|^2}{d_4(n)}\le\|f\|_{\mathscr{A}^1}^2 \tag{9}$$

Here $\zeta(s)^k=\sum_n d_k(n)n^{-s}$.

Lemma 5. *Suppose that (X,k) is $(\mathscr{H}^1,2)$ or $(\mathscr{A}^1,4)$, and that ℓ is a bounded linear functional on X with $\ell(f)=\sum_{n\ge 2}a_n/(\sqrt{n}(\log n)^{k/2})$ for Dirichlet polynomials f. Then for every $f\in X$ this series converges in its natural order, and its sum is $\ell(f)$.*

Proof. Write $w_n = 1/(\sqrt{n}(\log n)^{k/2})$. For $u \geq 1$, since $n^{1/(2u)} < e$ for $n < e^{2u}$ and $\zeta(1+x) \leq 1 + 1/x$,

$$\sum_{e^u \leq n < e^{2u}} \frac{d_k(n)}{n \log^k n} \leq \frac{e}{u^k} \sum_{n \geq 1} \frac{d_k(n)}{n^{1+1/(2u)}} = \frac{e\,\zeta(1+1/(2u))^k}{u^k} \leq e\,3^k.$$

Summing over the dyadic blocks of $\log n$ below and above $T \geq 1$, using $1 - e^{-x} \leq x$ and e^{-x} respectively, we find that

$$\sum_{n \geq 2} \frac{d_k(n)}{n \log^k n} \bigl| \mathbf{1}_{\{\log n \leq T\}} - e^{-(\log n)/T} \bigr|^2 \leq C$$

with C independent of T. Take $T = \log N$. By the Cauchy–Schwarz inequality and (8) or (9),

$$\sum_{2 \leq n \leq N} a_n w_n - \sum_{n \geq 2} a_n w_n n^{-1/\log N} \to 0 \qquad (N \to \infty).$$

Indeed, the terms with $n \leq M$ tend to zero for each fixed M, and the remaining terms contribute at most $C^{1/2} \bigl(\sum_{n > M} |a_n|^2 / d_k(n) \bigr)^{1/2}$, uniformly in N.

Next, the translation $f_\varepsilon(s) = f(s + \varepsilon)$ corresponds to the dilation $z_j \mapsto p_j^{-\varepsilon} z_j$ of the variables of the Bohr lift. Integral means of analytic functions over circles increase with the radius, so $f \mapsto f_\varepsilon$ is a contraction of X (for $\mathscr{A}^1$ this is also in [3, Theorem 11]), and by density of the polynomials $f_\varepsilon \to f$ in X as $\varepsilon \to 0^+$. For each $\varepsilon > 0$ the series $\sum_n a_n w_n n^{-\varepsilon}$ converges absolutely, by the Cauchy–Schwarz inequality and (8) or (9), and the same estimate applied to polynomial approximants shows that its sum is $\ell(f_\varepsilon)$. Its limit as $\varepsilon \to 0^+$ is therefore $\ell(f)$, and taking $\varepsilon = 1/\log N$ completes the proof. □

Proof of Corollary 1. For a Dirichlet polynomial f, the identity $\int_0^\infty n^{-\sigma}\,\mathrm{d}\sigma = 1/\log n$ gives

$$a_1 + \sum_{n \geq 2} \frac{a_n}{\sqrt{n} \log n} = \int_0^1 f(1/2 + \sigma)\,\mathrm{d}\sigma + \int_1^\infty \bigl(f(1/2+\sigma) - a_1 \bigr)\,\mathrm{d}\sigma.$$

The first integral is bounded by Theorem 2 with $\alpha = 1$ and $p = 1$. Integrating F against $\overline{z^{\kappa(n)}}$ gives $|a_n| \leq \|f\|_{\mathscr{H}^1}$, so the second integral is at most $\|f\|_{\mathscr{H}^1} \sum_{n \geq 2} n^{-3/2}/\log n$. Summing the terms with $n \leq 20$ and bounding the rest by $\int_{20}^\infty x^{-3/2}\,\mathrm{d}x/\log 20$ shows that this series is less than 1.164, and $5 + 2\sqrt{2} + 1.164 < 9$. The functional therefore extends continuously to $\mathscr{H}^1$. Since $f \mapsto a_1$ is bounded on $\mathscr{H}^1$, Lemma 5 with $k = 2$, applied to the functional minus a_1, shows that its value at every $f \in \mathscr{H}^1$ is given by the series. Since $\mathscr{H}^1$ is isometric to a subspace of $L^1(\mathbb{T}^\infty)$, the Hahn–Banach theorem represents the functional by a function $B \in L^\infty(\mathbb{T}^\infty)$ with the same norm, whose analytic projection is $1 + \sum_{n \geq 2} z^{\kappa(n)}/(\sqrt{n}\log n)$. □

Proof of Theorem 3. Let f be a Dirichlet polynomial. Since $\int_0^\infty n^{-\sigma}\sigma\,\mathrm{d}\sigma = 1/\log^2 n$, we have

$$\ell(f) = \sum_{n\ge 2} \frac{a_n}{\sqrt{n}\log^2 n} = \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\,\sigma\,\mathrm{d}\sigma.$$

On $(0,1)$ we use Theorem 2 with $\alpha = 2$ and $p = 1$, together with $|a_1| \le \|f\|_{\mathscr{A}^1}$. Integrating F against $\overline{z^{\kappa(n)}}$ gives $|a_n| \le d(n)\|f\|_{\mathscr{A}^1}$, since $\int |z|^{2k}\,\mathrm{d}m_2 = 1/(k+1)$. Hence the integral over $(1,\infty)$ is at most

$$\|f\|_{\mathscr{A}^1} \sum_{n\ge 2} \frac{d(n)}{n^{3/2}}\Big(\frac{1}{\log n} + \frac{1}{\log^2 n}\Big) \le \Big(\frac{1}{\log 2} + \frac{1}{\log^2 2}\Big)(\zeta(3/2)^2 - 1)\|f\|_{\mathscr{A}^1}$$
$$< 20.6\|f\|_{\mathscr{A}^1}.$$

Altogether $|\ell(f)| < (3.11 + 0.5 + 20.6)\|f\|_{\mathscr{A}^1} < 25\|f\|_{\mathscr{A}^1}$, so ℓ extends continuously to $\mathscr{A}^1$, and by Lemma 5 with $k = 4$ its value at every $f \in \mathscr{A}^1$ is given by the series. By the Hahn–Banach theorem ℓ is represented by some $\psi \in L^\infty(m_2^\infty)$ with $\|\psi\|_\infty \le 25$, whose moments are $\langle z^{\kappa(n)}, \psi\rangle = 1/(\sqrt{n}\log^2 n)$ for $n \ge 2$ and 0 for $n = 1$. Since $\|z^{\kappa(n)}\|^2_{L^2(m_2^\infty)} = 1/d(n)$, the analytic projection of ψ is φ. □

7. Measures on curves

The proof of Theorem 4 is most transparent in the logarithmic coordinate $x = \log(1/\sigma)$. We will see that $\|K_{r(\sigma)}\|_2^2$ is comparable to $\sigma^{-\kappa}$, and that the affinity between the points at σ and τ is bounded above by a constant times $e^{-\kappa|x-y|/4}$, with $y = \log(1/\tau)$. The condition $[\nu]_\kappa < \infty$ says exactly that the measure $\|K_{r(\sigma)}\|_2^2\,\mathrm{d}\nu(\sigma)$ has uniformly bounded mass on each interval $(e^{-k-1}, e^{-k}]$. So each scale carries bounded normalized mass and interacts with the other scales through a geometric series; this is the Schur test. For the converse, the extremal functions $K_{r(t)}^{2/p}$ of the Cole–Gamelin estimate are large on all of $(0,t]$.

In this section R has density $\theta \in (0,1]$ with constant B, and we write $r = r_R$. We work in the variables indexed by a finite subset R' of R, and we write

$$\|K_{r(\sigma)}\|_2^2 = \prod_{p\in R}(1-p^{-1-2\sigma})^{-\alpha}, \quad E\big(r(u),r(v)\big) = \sum_{p\in R}\rho(p^{-1/2-u}, p^{-1/2-v})^2$$

for the quantities over all of R. For real $a, b \in [0,1)$ each factor $((1-a^2)(1-b^2))^{-\alpha/4}(1-ab)^{-\alpha/2}$ in the bound of Lemma 3 is at least 1. Hence the kernel $\mathcal{G}$ in the variables R' is at most the product of these factors over all of R, which by (5) is at most

$$\|K_{r(\sigma)}\|_2\|K_{r(\tau)}\|_2 \exp\Big(-\frac{\alpha}{4}E\big(r(\sigma),r(\tau)\big)\Big).$$

All constants below are therefore independent of R'. We first record two consequences of (3).

Lemma 6. *Suppose that* $0 < u, v, \sigma \le 1$. *Then*

$$E(r(u), r(v)) \ge \theta|\log u - \log v| - \theta \log 4 - 4B$$

and

$$e^{-\alpha B}(2\sigma)^{-\kappa} \le \prod_{p\in R}(1-p^{-1-2\sigma})^{-\alpha} \le e^{\alpha(1+B)}(2\sigma)^{-\kappa}.$$

Proof. For $a, b \in [0,1)$ we have $\rho(a,b) \ge |a-b|$, so $E(r(u), r(v)) \ge S(2u) + S(2v) - 2S(u+v)$ with $S(t) = \sum_{p\in R} p^{-1-t}$. By (3) this is at least $\theta \log((u+v)^2/(4uv)) - 4B$, and $(u+v)^2/(uv) \ge e^{|\log u - \log v|}$. For the second claim, use $x \le -\log(1-x) \le x + x^2$ for $0 \le x \le 1/2$, the bound $\sum_p p^{-2} < 1$, and (3) with $t = 2\sigma$. □

Proof of Theorem 4. Clearly (2) implies (1). By Theorem 1, (4) implies (2), with the constant $\sup_{R'} \|\mathcal{G}_{\nu,R'}\|$ and polynomials in the variables R'.

(3) implies (4). We use (4) with the weight $w(\tau) = \|K_{r(\tau)}\|_2$. By the remark before Lemma 6, the row at σ is at most

$$\int \exp\Big(-\frac{\alpha}{4}E(r(\sigma), r(\tau))\Big)\|K_{r(\tau)}\|_2^2 \,\mathrm{d}\nu(\tau).$$

Let $I_k = (e^{-k-1}, e^{-k}]$ for $k \ge 0$, and let $\mathrm{d}\lambda(\tau) = \|K_{r(\tau)}\|_2^2 \,\mathrm{d}\nu(\tau)$. By Lemma 6, $\lambda(I_k) \lesssim e^{(k+1)\kappa}\nu(I_k) \lesssim [\nu]_\kappa$, and, since $|\log\sigma - \log\tau| \ge |j-k| - 1$ for $\sigma \in I_j$ and $\tau \in I_k$, the exponential is $\lesssim e^{-\kappa|j-k|/4}$ there. Hence the row at $\sigma \in I_j$ is $\lesssim \sum_{k\ge 0} e^{-\kappa|j-k|/4}\lambda(I_k) \lesssim [\nu]_\kappa$. Here and below, implied constants depend only on α, θ and B.

(1) implies (3). Suppose that (1) holds for some p with constant C. Testing with $F = 1$ shows that $\nu((0,1]) \le C$. Fix $0 < t \le 1$ and a finite $R' \subset R$, and let $F_t = K_{r(t)}^{2/p}\|K_{r(t)}\|_2^{-2/p}$, computed in the variables R'. Then $\|F_t\|_p = 1$, and F_t is holomorphic in a neighbourhood of the closed polydisc, so its Taylor polynomials converge to it uniformly there; since ν is finite, (1) holds for F_t. For $\sigma \le t$ we have $K_{r(t)}(r(\sigma)) = \prod(1-p^{-1-\sigma-t})^{-\alpha} \ge \|K_{r(t)}\|_2^2$, both computed in the variables R', so $|F_t(r(\sigma))|^p \ge \|K_{r(t)}\|_2^2$. Letting R' increase to R, Lemma 6 gives $\nu((0,t])e^{-\alpha B}(2t)^{-\kappa} \le C$, that is, $[\nu]_\kappa \le 2^\kappa e^{\alpha B}C$. Together with the proof that (3) implies (4), this shows that the best constant in (1) is comparable to $[\nu]_\kappa$. □

Theorem 4 also applies to atomic measures. For every choice of $\tau_k \in \overline{I_k} = [e^{-k-1}, e^{-k}]$, the measure $\nu = \sum_{k\ge 0} e^{-k\kappa}\delta_{\tau_k}$ satisfies $[\nu]_\kappa \le e^\kappa/(1-e^{-\kappa})$, and hence $\sum_k e^{-k\kappa}|F(r(\tau_k))|^p \le C\|F\|_p^p$ with C independent of the choice, for instance when τ_k maximizes $|F(r(\cdot))|$ on $\overline{I_k}$.

8. Concluding remarks

The constant near the critical line. Near the origin, the kernel (6) for $\alpha = 1$ is asymptotic to $L_0(\sigma,\tau) = 2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$, whose norm on $L^2(0,\infty)$ is $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$. Since $\zeta(1+x) \ge 1/x$, the kernel (6) dominates L_0, which by homogeneity has the same norm on every interval

$(0,\delta)$. So the bound (6) cannot give a constant below $\Gamma(1/4)^2/\sqrt{2\pi}$ on any interval $(0,\delta)$. This concerns the method, not the best constant in (2). That constant is at least π: for $p=2$ the kernel $\langle K_{r(\sigma)}, K_{r(\tau)}\rangle$ characterizes the embedding exactly and dominates $1/(\sigma+\tau)$, whose norm on $L^2(0,1)$ is π, and applying (2) to $f=g^2$ gives the case $p=2$ with the same constant.

Problem 1. *Determine the best constant C in $\int_0^1 |f(1/2+\sigma)|\,\mathrm{d}\sigma \le C\|f\|_{\mathscr{H}^1}$.*

What the Gram condition loses. Theorem 4 shows that the condition of Theorem 1 is necessary on the curves r_R for sets of primes R with a density. It is not necessary in general. In one variable with $\alpha=1$, let m_r be normalized arc length on the circle $|z|=r$, and let $\mu=\sum_{j\ge1} 2^{-j} m_{r_j}$ with $r_j = 1-e^{-2^j}$. Subharmonicity gives $\int |F|^p\,\mathrm{d}\mu \le \|F\|_p^p$ for every p. On the other hand, the proof of Theorem 1 shows that $\|\mathcal{G}_\mu\| \ge \int \langle u, |K_z|\rangle^2\,\mathrm{d}\mu(z)$ for every unit vector $u\in L^2(m)$. With $u=1$, since $\langle 1, |K_z|\rangle$ is of order $\log(1/(1-|z|))$, the right-hand side is of order $\sum_j 2^{-j}4^j=\infty$, so $\mathcal{G}_\mu$ is unbounded. The loss comes from directions in which the kernels rotate: $K_{e^{i\theta}r}$ and K_r differ by a rotation, and the cancellation which makes circle averages harmless is invisible to $|K|$.

Other settings. The proofs use only Lemma 1, the subharmonicity of $|H|^q$ for holomorphic H, and Lemma 3. Hence they apply verbatim to analytic polynomials with values in a Banach space X, with $\|F\|_X$ in place of $|F|$. They also apply to the Gaussian measure $\pi^{-d}e^{-|z|^2}\,\mathrm{d}V(z)$ on $\mathbb{C}^d$, where V is Lebesgue measure (the analytic closure is then the Fock space), with $K_r(z)=e^{\overline{r}\cdot z}$ and $T_r(w)=r+w$; there $\langle h_r, h_s\rangle = e^{-|r-s|^2/4}$ exactly.

The Hardy space question of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt. Immediately before [3, Theorem 16] the authors ask whether φ defines a bounded functional on the Hardy space $H^1(\mathbb{D}^\infty)$, where the pairing is $\sum_n a_n d(n)/(\sqrt{n}\log^2 n)$. Theorem 3 does not answer that question. By (9), the absolute series $\sum |a_n|/(\sqrt{n}(\log n)^\gamma)$ is bounded by a constant times $\|f\|_{\mathscr{A}^1}$ for $\gamma>2$.

Problem 2. *Is $\sum_{n\ge2}|a_n|/(\sqrt{n}\log^2 n) \le C\|f\|_{\mathscr{A}^1}$ for every $f\in\mathscr{A}^1$?*

Acknowledgements. AI assistance was used in developing and checking the arguments and in preparing the exposition; the author has verified the proofs.

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Email address: `oistein.sovik@usit.uio.no`